7e. Functions: Type Inference and Multiple Dispatch

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INTRODUCTION

In Julia, functions are key for achieving high performance. This is by design, as they have been engineered from the outset to generate efficient machine code. In fact, the main practical implication from this section will be that **wrapping code in functions is crucial for high performance in Julia**.

However, to fully unlock the potential of functions, we must first understand the underlying process of function calls. Essentially, when a function is called, Julia attempts to identify concrete types for its variables, eventually selecting a corresponding computation method. At the heart of the process are three interconnected mechanisms: **dispatch**, **compilation**, and **type inference**. This section will provide a detailed explanation of each concept.

VARIABLE SCOPE IN FUNCTIONS

To understand why functions are central to writing high-performance code, it helps to revisit the variable scope of functions. Recall that **local variables** include its arguments and any variable created within the function body. These variables exist only during the function's execution and are inaccessible from outside. **Global variables**, on the other hand, refer to any variable defined outside the function and remain accessible throughout program execution.

When code runs inside a function and all its variables are local, the compiler can reason about types, optimize aggressively, and generate efficient machine code. Global variables disrupt this process, as their types and values can change at any time, forcing the compiler to insert dynamic lookups and prevents key optimizations. The result is a substantial performance degradation.

Note that the use of global variables isn't limited to code written in the global scope. It also arises when a function references or writes variables that aren't passed as arguments. As a result, the same performance drawbacks from global variables will appear in all the following cases.

GLOBAL SCOPE

```
x   = 2

y   = 3 * x
```

```
julia> y
6
```

FUNCTION USING A GLOBAL VARIABLE

```
x   = 2

f() = 3 * x
```

```
julia> f()
6
```

Recall that an assignment like `x = 2` is shorthand for `x::Any = 2`, reflecting that global variables default to `Any` if they aren't explicitly type-annotated. Furthermore, only concrete types can be instantiated, meaning that values can only adopt a concrete type. This is why `x::Any` shouldn't be interpreted as `x` having type `Any`, but rather that `x` can take on any concrete type that's a subtype of `Any`. Since `Any` sits at the top of Julia's type hierarchy, this simply means that `x`'s types are unrestricted.

This unrestricted nature of global variables is precisely what prevents Julia from specializing operations. In our example, `x` is a global variable, and the specialization of `*` is therefore precluded. The issue arises because, once a global variable is introduced, Julia must consider multiple possible methods for the computation of `*`, one for each possible concrete type of `x`. In practice, this leads to code generation with multiple branches, potentially involving type checks, conversions, and memory allocations. The consequence is degraded performance.

The underlying logic for Julia treating global variables as potentially embodying any type and value is that, even if a variable holds some value at a specific moment, the user may reassign it at any point in the program. Note that the performance issues wouldn't completely go away if we had type-annotated `x` with a concrete type such as `x::Int64 = 2`. Many optimizations depend not only on knowing the concrete type of a variable, but also on knowing its value and on having a well-defined scope in which that value exists. It's only when these assumptions are met that Julia can gain a comprehensive view of all the operations to be performed, creating opportunities for optimizations.

Functions in Julia are designed precisely to provide these guarantees. By enforcing local scope and predictable variable behavior, they enable method specialization and unlock the full range of performance optimizations. Next, we explore the specific steps that Julia takes to achieve this.

FUNCTIONS AND METHODS

A **function** is simply a name that can be associated with different implementations. Each implementation is known as a **method**, which specifies the function body for a particular combination of argument types and number of arguments. You can inspect the methods associated with a function `foo` by executing `methods(foo)`.

To see this in action, let's define several methods for the function `foo1`. Creating a method requires type-annotating the arguments of `foo1` during its definition, which is implemented via the `::` operator. In this way, we can set a distinct function body for each unique combination of argument types.

To keep matters simple, let's begin with a scenario where all the methods of `foo1` take the same number of arguments, thus differing only in their types.

METHODS

```
foo1(a,b)                    = a + b
foo1(a::String, b::String) = "This is $a and this is $b"
```

```
julia> methods(foo1)
2 methods for generic function "foo1" from Main
julia> foo1(1,2)
3
julia> foo1("some text", "more text")
"This is some text and this is more text"
```

Since `foo1(a,b)` is equivalent to `foo1(a::Any,b::Any)`, the first method sets the behavior of `foo1` for every possible pair of argument types. After this, when we introduce the method `foo1(a::String, b::String)`, we override that default for the specific case in which both arguments are strings. The existence of multiple methods explains why the two calls produce different outputs: the first method of `foo1` is called with `foo1(1, 2)`, whereas `foo1("some text", "more text")` triggers the second method.

The example also reveals that **methods don't need to perform similar operations**. Although it's usually unwise to group unrelated behaviors under a single function name, the ability to tailor implementations is central to performance-oriented programming. In practice, developers often exploit this flexibility to provide optimized algorithms for particular types, ensuring that a function can adapt its behavior while maintaining a coherent interface.

Also, **methods don't need to have the same number of arguments**. For instance, it's possible to define the following methods for a function `foo2`.

METHODS WITH DIFFERENT NUMBERS OF ARGUMENTS

```
foo2(x)       = x
foo2(x, y)    = x + y
foo2(x, y, z) = x + y + z
```

```
julia> methods(foo2)
3 methods for generic function "foo2" from Main
julia> foo2(1)
1
julia> foo2(1, 2)
3
julia> foo2(1, 2, 3)
6
```

Defining methods with different number of arguments is particularly useful for extending a function's behavior. A prime example is given by the function `sum`. So far, we've only used its simplest form `sum(x)`, which adds all the elements of a collection `x`. However, `sum` also supports additional methods. One of them is `sum(<function>, x)`, where the elements of `x` are transformed via `<function>` before being summed.

METHODS FOR 'SUM'

```
x = [2, 3, 4]

y = sum(x)          # 2 + 3 + 4
z = sum(log, x)     # log(2) + log(3) + log(4)
```

FUNCTION CALLS

Building on our understanding of how functions and methods are defined, let's now analyze the process triggered when a function is called. All our explanations will be based on the following function `foo`:

EXAMPLE

```
foo(a, b) = 2 + a * b
```

```
julia> foo(1, 2)
4
julia> foo(3, 2)
8
julia> foo(3.0, 2)
8.0
```

Defining a function like `foo(a, b)` is shorthand for creating a **method** with the signature `foo(a::Any, b::Any)`. Thus, the function body `foo(a,b)` holds for all possible type combinations of `a` and `b`.

When `foo(1, 2)` is called, Julia evaluates the expression `2 + a * b` through a series of steps.

The process begins with what's known as **multiple dispatch**, where Julia selects which method of the function to execute. Importantly, this decision is based solely on the concrete types of the function arguments, not their values.

Specifically in our example, `a = 1` and `b = 2` are identified as `Int64`. The information on types is then used to select a *method*, which defines the function body and hence the operations to be performed. This process involves searching through the available methods of `foo`, eventually choosing the most specific one that matches the concrete types of `a` and `b`. Since `foo` has only one method `foo(a,b) = 2 + a * b`, this applies to any argument types, including `a::Int64` and `b::Int64`. Therefore, the corresponding function body is `2 + a * b`.

Defining a function like `foo(a, b)` is shorthand for creating a **method** with the signature `foo(a::Any, b::Any)`. This means the function body applies to every possible combination of argument types. When we call `foo(1, 2)`, Julia evaluates the expression `2 + a * b` by following a well-defined sequence of steps.

The process begins with **multiple dispatch**, Julia's mechanism for selecting which method of a function to run. Crucially, this selection depends only on the *types* of the arguments, not their values. Julia first determines the concrete types of the inputs. In our example, both `a = 1` and `b = 2` are identified as `Int64`. With these types in hand, Julia searches through all methods of `foo` to find the most specific one that matches the pair `(Int64, Int64)`.

In this case, `foo` has only one method—`foo(a, b) = 2 + a * b`—which is defined for all argument types, including `Int64`. As a result, Julia selects this method and uses its function body.

Once the method is chosen, Julia looks for a **method instance**, which is the compiled version of the method specialized for the signature `foo(a::Int64, b::Int64)`. If such an instance already exists, Julia reuses it immediately to compute `foo(1,2)`. If not, the compiler generates optimized machine code for that specific type combination, caches it for future calls, and then executes it. This on-demand specialization is one of the key reasons Julia can offer performance close to low-level languages.

The following diagram depicts the process unfolded when `foo(1,2)` is executed.

MULTIPLE DISPATCH

The process outlined has implications for how the language works: the first time a function is called with a particular combination of concrete types, Julia incurs a time cost to generate specialized code for those types. This initial delay is often referred to as Time To First Plot, a phrase that highlights how the compilation overhead becomes noticeable in plotting libraries.

Note, though, that once Julia compiles a method instance, it caches the resulting code. Consequently, any later call with the same argument types can reuse that cached instance immediately, avoiding the compilation step entirely. In practical terms, it implies that all the subsequent function calls with the same concrete types are fast.

The behavior of `foo` makes this clear. After evaluating `foo(1, 2)`, Julia has already compiled a method instance for the signature `foo(a::Int64, b::Int64)`. Consequently, the subsequent call `foo(3, 2)` is executed immediately by invoking the cached method instance, without any need for recompilation. Instead, the execution of `foo(3.0, 2)` introduces a new combination of argument types for `a::Float64` and `b::Int64`. Because no compiled method instance yet exists for this signature, Julia must generate one before executing the function.

TYPE INFERENCE

As we indicated, Julia produces the concrete machine code that will ultimately run the computation during compilation. This compilation in Julia happens on demand, right at the moment a function is first called with a new set of argument types. The strategy is known as **Just-In-Time Compilation (JIT)**.

Most considerations for achieving high performance are tied to the compilation process. A key mechanism in it is **type inference**, whereby the compiler attempts to identify concrete types for *all* variables and expressions within the function body.

If the compiler succeeds in identifying concrete types, it can specialize instructions for each operation and yield fast code. For instance, type inference with the function `foo` defined above involves determining concrete types for `2`, `a = 1`, and `b = 2`. Since all values have type `Int64`, the compiler can specialize the computation of `2 + a * b` for variables with type `Int64`. This is the essence behind **type stability**, which we'll cover extensively in the next chapter.

On the contrary, if the compiler is unable to identify concrete types for some expressions, it must create generic code to accommodate multiple combinations of types. This forces Julia to perform type checks and conversions during runtime, significantly degrading performance.

Below, we provide various remarks about type inference that are worth keeping in mind for next sections.

FUNCTIONS DO NOT GUARANTEE THE IDENTIFICATION OF CONCRETE TYPES

Merely wrapping code in a function doesn't guarantee that the compiler will identify concrete types for all operations. The following example illustrates this.

TYPE-UNSTABLE FUNCTION

```
x       = [1, 2, "hello"]     # Vector{Any}

foo3(x) = x[1] + x[2]         # type unstable
```

```
julia> foo3(x)
3
```

The issue in the example is that the compiler assigns the type `Any` to both `x[1]` and `x[2]`, as they come from an object with type `Vector{Any}`. As a consequence, the compiler can't specialize the computation of the operation `+`. The example also highlights that compilation is exclusively based on types, not values. Thus, the generated code ignores the actual values `x[1] = 1` and `x[2] = 2`, which would otherwise reveal the type `Int64`.

GLOBAL VARIABLES INHERIT THEIR GLOBAL TYPE

Type inference is restricted to local variable. Instead, any global variable inherits its type from the global scope. In the following examples, `b` and `d` are global variables. Consequently, the compiler defines `b`'s type as `Any` and `d` as `Number`.

UNANNOTATED GLOBAL VARIABLE

```
a         = 2
b         = 1

foo4(a)   = a * b
```

```
julia> foo4(a)
2
```

TYPE-ANNOTATED GLOBAL VARIABLE

```
c         = 2
d::Number = 1

foo4(c)   = c * d
```

```
julia> foo4(c)
2
```

TYPE-ANNOTATING FUNCTION ARGUMENTS DOES NOT IMPROVE PERFORMANCE

We pointed out that identifying concrete types is crucial for achieving high performance. This might wrongly suggest that explicitly annotating function arguments is necessary for performance, or at least beneficial. Such annotations are actually redundant, thanks to type inference. In fact, adding them can be counterproductive, as they unnecessarily restrict the types accepted by a function, thereby limiting its flexibility and potential applications.

To see how this loss of flexibility plays out, compare the following scripts.

UNANNOTATED FUNCTION

```
foo5(a, b) = a * b
```

```
julia> foo5(0.5, 2.0)
1.0
julia> foo5(1, 2)
2
```

TYPE-ANNOTATED FUNCTION

```
foo6(a::Float64, b::Float64) = a * b
```

```
julia> foo6(0.5, 2.0)
1.0
julia> foo6(1, 2)
ERROR: MethodError: no method matching foo6(::Int64, ::Int64)
```

The function on the first tab only accepts arguments of type `Float64`, implying that even integers are disallowed. By contrast, the function on the second tab also accepts the same behavior for `Float64` inputs, but additionally allows for other types, due to implicit type annotation `Any` on the function arguments.

Packages Commonly Type-Annotate Function Arguments

When inspecting code of packages, you may notice that function arguments are often type-annotated. The reason for this isn't related to performance, but rather to ensure the function's intended usage, safeguarding against inadvertent type mismatches.

For instance, suppose a function that computes the revenue of a theater via `nr_tickets * price`. Importantly, the operator `*` in Julia not only implements product of numbers, but also concatenates words when applied to expressions with type `String`. Without type-annotations, the function could potentially be misused, as exemplified in the first tab below. The second tab precludes this possibility by asserting types.

UNANNOTATED FUNCTION

```
revenue1(nr_tickets, price) = nr_tickets * price
```

```
julia> revenue1(3, 2)
6
julia> revenue1("this is ", "allowed")
"this is allowed"
```

TYPE-ANNOTATED FUNCTION

```
revenue2(nr_tickets::Int64, price::Number) = nr_tickets * price
```

```
julia> revenue2(3, 2)
6
julia> revenue2("this is ", "not allowed")
ERROR: MethodError: no method matching revenue2(::String,
::String)
```